Homework 4: My answers :)

PART A

In OTC trades, collateral is a deposit against a payoff that still occurs at maturity. The contract keeps a non-zero value equal to that future cashflow seen from today so it must be discounted. The collateral is returned with interest, which is what discounting at the collateral rate represents. In futures, each daily variation margin is a final settlement. The money exchanges hands for good, no interest is owed on it, and the contract's value resets to 0 every day. Because each new position costs 0 to enter, the pricing condition is $0 = D(\text{expected change in } F)$. D is known & fixed so it cancels. The future price is therefore not a discounted value but a martingale.

$$F(t) = E^{Q}[S(T)]$$

PART B

Building an interest-rate curve from a list of rates, use it to price one future payment, and work out how sensitive that price is to each rate.

computed on github :)

https://github.com/SM-dot/FRE-GY-9743-Assignments-4/tree/assignment-4-submission